

Systematic Review

Machine Learning and Generative AI in Learning Analytics for Higher Education: A Systematic Review of Models, Trends, and Challenges

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Featured Application

This review supports the design of hybrid learning analytics systems that combine ML and GenAI to enable early risk detection and personalized feedback in higher education.

Abstract

This systematic review examines how machine learning (ML) and generative AI (GenAI) have been integrated into learning analytics (LA) in higher education (2018–2025). Following PRISMA 2020, we screened 9590 records and included 101 English-language, peer-reviewed empirical studies that applied ML or GenAI within LA contexts. Records came from 12 databases (last search 15 March 2025), and the results were synthesized via thematic clustering. ML approaches dominate LA tasks, such as engagement prediction, dropout-risk modelling, and academic-performance forecasting, whereas GenAI—mainly transformer models like GPT-4 and BERT—is emerging in real-time feedback, adaptive learning, and sentiment analysis. Studies spanned world regions. Most ML papers ($n = 75$) examined engagement or dropout, while GenAI papers ($n = 26$) focused on adaptive feedback and sentiment analysis. No formal risk-of-bias assessment was conducted due to heterogeneity. While ML methods are well-established, GenAI applications remain experimental and face challenges related to transparency, pedagogical grounding, and implementation feasibility. This review offers a comparative synthesis of paradigms and outlines future directions for responsible, inclusive, theory-informed AI use in education.

Keywords: AI-enhanced learning; generative artificial intelligence; higher education; learning analytics; machine learning



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1. Introduction

Amid the growing integration of AI in education, machine learning (ML) and generative AI (GenAI) have gained prominence for their ability to personalize learning, predict outcomes, and support institutional decision-making [1–3]. While higher education fosters critical thinking and innovation [4], it continues to face challenges such as dropout, inequality, and the demand for adaptive environments [5]. In response, learning analytics (LA), enhanced by ML, is increasingly used to promote equity and student success through data-driven practices [6,7].

Defined by Siemens [8] as the analysis of data about learners and their contexts to optimize learning, LA has embraced ML techniques—supervised, unsupervised, and

semi-supervised—to forecast performance, detect disengagement, and enable timely interventions [9–12]. Recent advances in GenAI, especially large language models (LLMs), are expanding LA’s capabilities through automated feedback, intelligent tutoring, and interaction analysis [13–15], although empirical applications remain limited and underexplored in terms of pedagogy and ethics. Some studies caution that GenAI’s “black-box” nature and lack of pedagogical grounding may undermine trust and educational validity, contrasting with perspectives that emphasize its potential to enhance learner engagement and scalability.

Existing reviews on ML in LA [16–19] rarely address GenAI or offer an integrated synthesis of both approaches. To fill this gap, this review analyzes empirical studies from 2018 to 2025, including recent developments from major conferences.

The study addresses two research questions:

RQ1: How are ML and GenAI applied in LA within higher education?

RQ2: What benefits arise from their integration in this context?

The results indicate that, while ML remains dominant in predictive analytics, GenAI is emerging in real-time feedback and adaptive learning but still faces critical challenges around transparency and pedagogical validity. By examining methodological trends, educational applications, and broader implications, this review offers a timely perspective to guide the responsible adoption of AI in higher education.

2. Materials and Methods

2.1. Study Eligibility Criteria

This review adheres to PRISMA 2020 guidelines [20] to ensure methodological transparency and rigor. A completed PRISMA 2020 checklist is provided as Supplementary Materials. It synthesizes empirical studies on the use of ML and GenAI in learning analytics (LA) within higher education, applying stream-specific inclusion and exclusion criteria as follows:

- Inclusion Criteria (both streams);
 - Language: Articles must be published in English;
 - Accessibility: Full-text availability is required;
 - Study type: Only empirical studies with a clearly stated research question;
 - Context: The study must explicitly address a learning analytics objective in higher education.
- For LA + ML (2018–2023);
 - Must apply Machine Learning techniques within LA;
 - Must report empirical results, such as predictive performance or analytics-based outcomes.
- For LA + GenAI (2020–2025);
 - Must apply generative AI models in LA;
 - Applications must involve generation, adaptation, or feedback, aligned with LA goals.
- Exclusion Criteria.
 - Non-empirical works;
 - Preprints and non-peer-reviewed documents;
 - Studies lacking either an LA objective or the use of ML/GenAI;
 - Conference papers were not formally included in the final dataset to ensure consistent peer-review standards and methodological rigor across all selected studies. This decision aimed to maintain a homogeneous level of academic scrutiny, focusing exclusively on peer-reviewed journal articles.

This stream-specific distinction ensures a consistent comparative analysis and captures the emergence of GenAI as a distinct methodological paradigm within LA.

2.2. Data Sources

The literature search was conducted in two phases to reflect the evolution of the research questions and the emergence of generative AI (GenAI) in education.

In the first phase, covering ML-based learning analytics (LA + ML) from 2018 to 2023, 11 scholarly databases were consulted to construct a comprehensive and systematic corpus: ACM Digital Library, IEEE Xplore, Emerald, ERIC, ProQuest, Sage Journals, Web of Science, ScienceDirect, Wiley Online Library, Taylor & Francis, and Scopus.

In a subsequent phase, an updated search strategy was implemented to include studies on generative AI in learning analytics (LA + GenAI), published between 2020 and early 2025. Given the novelty and rapid development of GenAI applications, the search focused on sources with high coverage of emerging AI research: ACM Digital Library, IEEE Xplore, SpringerLink (EC-TEL), and Scopus. This approach responded to the more limited but rapidly evolving body of GenAI literature in education, which tends to be concentrated in fewer venues.

While the formal PRISMA-based corpus was limited to peer-reviewed journal articles, an exploratory review of flagship conference proceedings, including LAK, L@S, and EC-TEL, was conducted for the full period (2018–2025) to identify recent trends not yet indexed. Selected contributions, particularly from 2024 and 2025, are discussed in the Related Work section to enrich the contextual interpretation.

2.3. Search Strategy

Search queries combined three conceptual pillars—learning analytics (including social learning analytics), machine learning or GenAI, and higher education. Synonyms and Boolean operators ensured comprehensive yet focused retrieval. The search spanned 2018–2025 (2018–2023 for ML in LA). Full queries are detailed in the Supplementary Materials (Table S1) and available on Zenodo (<https://doi.org/10.5281/zenodo.15233231>). Of the 9590 records retrieved, 8263 were excluded after deduplication and screening (see PRISMA flowchart, Figure 1).

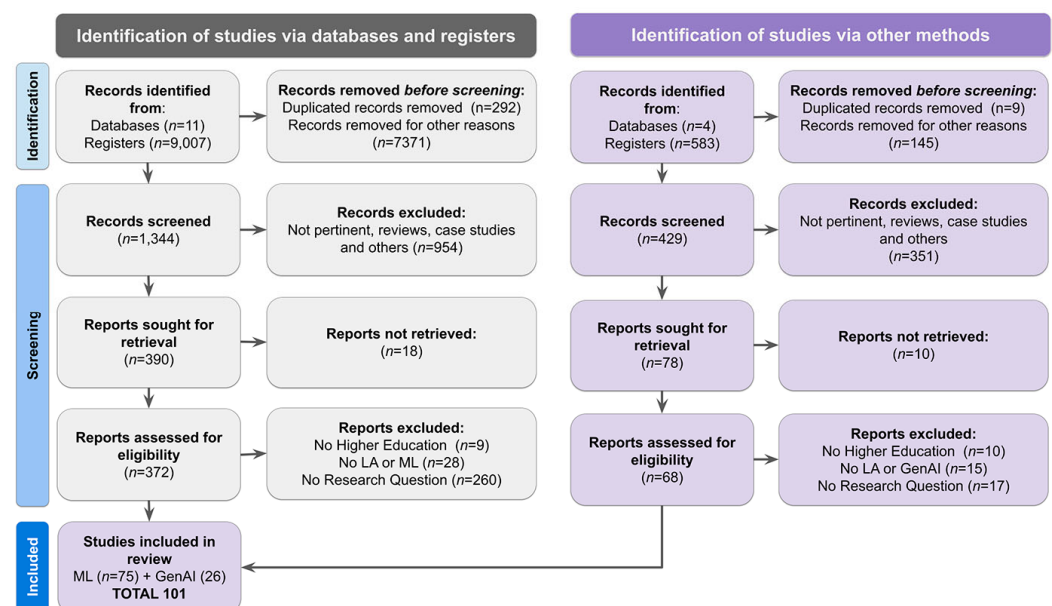


Figure 1. PRISMA flowchart for study selection (2018–2025), detailing inclusion of 101 studies from an initial pool of 9590 records.

2.4. Data Extraction and Collection Process

All references were managed using RefWorks ProQuest. After deduplication, two parallel screening processes were conducted. For LA + ML, 1344 records were screened, 954 excluded, and 372 assessed for eligibility, yielding 75 empirical studies. For LA + GenAI, 429 records were screened, 351 excluded due to irrelevance or lack of empirical content, and 68 assessed, resulting in 26 included studies. Figure 1 summarizes the inclusion and exclusion process.

Screening was conducted using a structured coding protocol aligned with the review's research questions. All screening stages (titles/abstracts and full texts) were performed by a single reviewer. Each record was evaluated against the predefined eligibility criteria, with particular emphasis on the presence of an explicit research question and the study's relevance to addressing the review objectives.

2.5. Quality Appraisal

To assess the methodological quality of the included studies, we applied the Mixed Methods Appraisal Tool (MMAT, 2018 version) [21]. Each study was classified according to MMAT study type (qualitative, quantitative, mixed methods) and assessed using its corresponding checklist. Screening criteria (S1 and S2) and methodological dimensions (e.g., sampling, measurement, data analysis) were evaluated and summarized. The complete dataset with MMAT ratings, checklist responses, study types, and reviewer notes for all 101 studies is openly available on Zenodo (<https://doi.org/10.5281/zenodo.16416487>). This open dataset enhances transparency and supports reproducibility of the review process.

2.6. Final Dataset

The final review includes 101 empirical studies: 75 focused on traditional ML applications in learning analytics and 26 integrating GenAI models, such as GPT, BERT, or FLAVA, into LA contexts. This dual corpus enables analysis of both the evolution of ML and the rise of GenAI within the LA research landscape.

2.7. Declaration of GenAI Use

During the study, ChatGPT-4.5 (OpenAI) was used to improve language clarity and to assist in refining the Python 3.10 code. All AI-generated output was reviewed and edited by the authors, who take full responsibility for the final content.

2.8. Data Availability

The structured dataset (metadata extracted from 101 studies, including AI models, contexts, and techniques) is publicly available via Zenodo (<https://doi.org/10.5281/zenodo.16416465>) to ensure transparency and reproducibility.

2.9. Ethical Considerations

No human or animal subjects were involved; ethical approval was not required.

2.10. Registration and Protocol

This review was not prospectively registered, and no separate protocol was prepared. Consequently, no protocol amendments apply.

3. Results

To structure the results, the corpus was divided into two groups—LA&ML (traditional machine learning) and LA&GenAI (generative AI)—based on each study's core methodology. Within these groups, unsupervised clustering was applied to identify thematic patterns using three categorical variables: AI models, application, and educational context, encoded

via one-hot encoding. Dimensionality reduction and visualization were performed with principal component analysis (PCA), and the optimal number of clusters was determined using silhouette scores. The analysis, conducted in Python via Google Colab, informed the thematic organization of Sections 3.2 and 3.3, enabling a data-driven presentation of the findings across both paradigms.

3.1. Temporal and Geographical Distribution of Publications

This section analyzes the evolution of learning analytics (LA) research using machine learning (ML) and generative AI (GenAI) in higher education between 2018 and 2025, focusing on temporal trends and geographical spread.

The annual trends (Figure 2) show consistent growth in ML-based LA studies, with a peak in 2022. GenAI studies first appeared in 2022 and accelerated in 2024, coinciding with the widespread release and adoption of large language models (LLMs) such as GPT-4. This pattern reflects two coexisting trajectories: the consolidation of traditional ML approaches—classification, regression, and ensemble models applied to LMS data—and the experimental rise of GenAI applications for feedback generation, engagement modeling, and content personalization. While ML remains foundational, GenAI is rapidly gaining traction, though it is still in the early stages of methodological standardization and empirical validation.

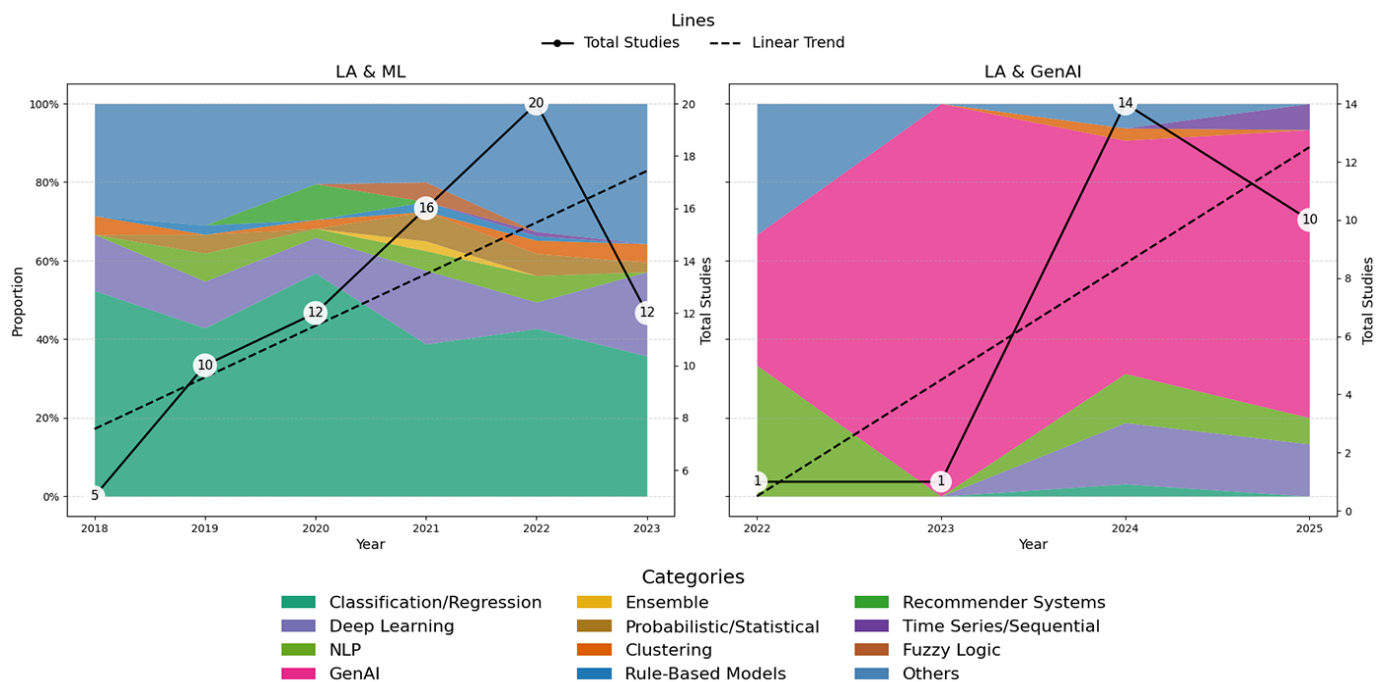


Figure 2. Temporal trends in learning analytics studies using traditional machine learning (left) and generative AI (right) in higher education between 2018 and 2025.

Geographical distribution (Figure 3) reveals notable disparities. The United States leads with 17 studies (10 ML, 7 GenAI), followed by Australia, China, and Germany, which show relatively balanced activity across both paradigms. However, GenAI research remains highly concentrated in high-income regions, with limited representation from Latin America, Sub-Saharan Africa, and parts of Asia. These gaps raise concerns about equity, contextual relevance, and disparities in access to GenAI infrastructure for research and implementation in education.

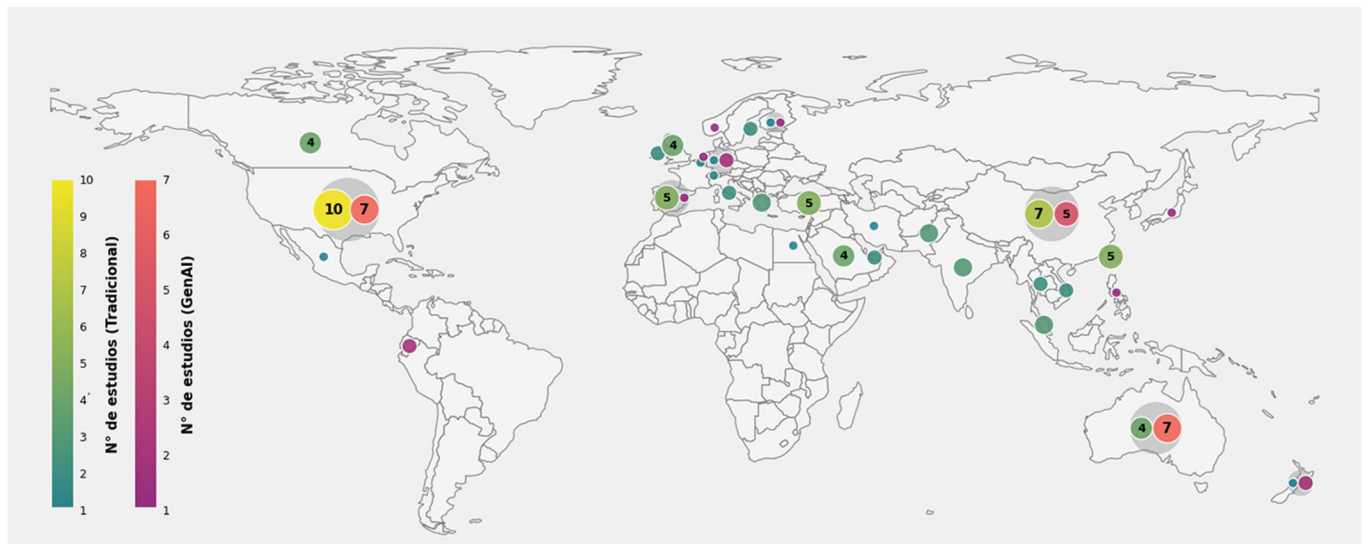


Figure 3. Geographical distribution of learning analytics studies in higher education (2018–2025) by AI model type.

The coexistence of ML and GenAI suggests a transitional phase in LA research. ML offers robustness and interpretability, while GenAI brings adaptability and personalization. Hybrid approaches could combine their strengths to enhance flexibility and inclusiveness.

Three key observations emerge: (1) GenAI adoption aligns with the release of tools like GPT-4, highlighting the role of accessibility; (2) its concentration in high-income regions may deepen epistemic inequalities; and (3) the parallel rise of both paradigms invites integrated, transparent, and context-aware frameworks.

3.2. Learning Analytics with Traditional Models

To analyze the application of traditional machine learning (ML) in learning analytics (LA), we clustered 75 peer-reviewed studies from 2018 to 2023. Using three categorical variables—AI models, application type, and educational context—encoded and reduced via principal component analysis (PCA), we identified thematic groupings centered on engagement prediction, dropout modeling, academic performance forecasting, and feedback systems across online, blended, and face-to-face settings.

Figure 4 presents the top 10 ML models by context. Random forest, support vector machine (SVM), and decision tree are most prevalent, particularly in online and MOOC environments, due to their robustness, interpretability, and compatibility with structured behavioral data. Logistic regression, naive Bayes, and artificial neural networks also appear frequently, indicating a reliance on established methods.

3.2.1. Engagement Prediction in Online Learning with Traditional ML Models

The analysis of student engagement in online and hybrid environments has progressed through the integration of ML within LA. This cluster, comprising 11 studies, focuses on predicting engagement, academic performance, and dropout risk using structured and unstructured data sources.

Most studies rely on classical ML algorithms, such as random forest, decision tree, SVM, and logistic regression, often enhanced with ensemble techniques like AdaBoost and Bagging. These models are widely used in MOOCs and online courses to support an early detection of at-risk behavior. However, few studies, including those addressing engagement and behavioral prediction [22–26], assess whether such predictions lead to real-time, actionable interventions.

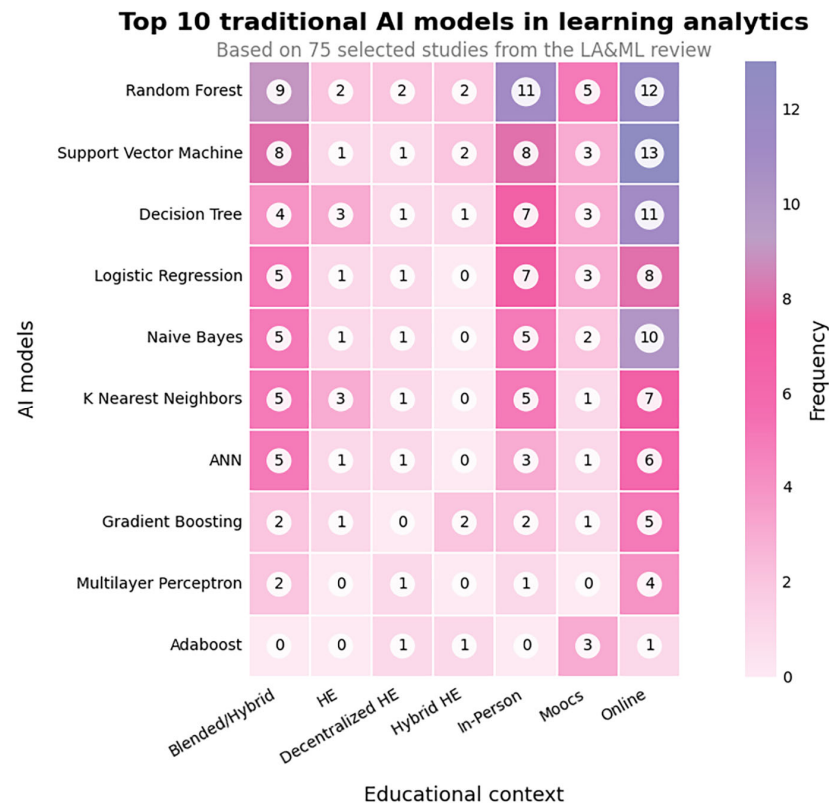


Figure 4. Top 10 AI models applied in learning analytics using traditional machine learning techniques, categorized by educational context.

The choice of algorithm often reflects the nature of the data: SVM and artificial neural networks (ANN) are used for fine-grained engagement prediction [24], while decision tree and logistic regression are suited to interpret behavioral traces such as video interaction logs [25]. Feature engineering, clustering, and regression remain central analytical techniques [27].

A notable methodological trend is the use of natural language processing (NLP) to analyze forum content and course reviews. Word embeddings—Doc2Vec, Word2Vec, Fast-Text, and GloVe—support sentiment classification and behavioral inference [28,29]. Deep learning models, such as LSTM and GRU, enhance performance in text-based engagement tasks, with GloVe consistently yielding superior accuracy [28].

Among the most comprehensive approaches is the work of Onan [28], which combines traditional classifiers, ensemble techniques, and deep architectures like CNN, GRU, LSTM, and attention-based RNN. The model achieves 95.80% accuracy using LSTM with GloVe embeddings, demonstrating the synergy between affective analytics and attention mechanisms—an intersection between traditional ML and GenAI.

Emerging studies also explore spatiotemporal learning behaviors. Du et al. [30] show that consistent study patterns correlate with academic success. Moubayed et al. [31] apply clustering to blended learning profiles, while Lahza et al. [32] investigate strategy use in learner sourcing platforms, linking it to instructional design quality.

Key applications include:

- Dropout detection and risk assessment [22,25];
- Engagement forecasting and behavioral profiling [24,26,27];
- Sentiment analysis of learner feedback [28];
- Modeling learning routines and platform interactions [30–32].

Challenges and opportunities: Multimodal integration remains limited, with most studies focusing on a single data type. Longitudinal analyses are rare, limiting insights into lasting engagement patterns. Although XAI is occasionally applied [23,26], most models lack transparency, reducing their practical utility in educational settings.

Recommendations for future research:

- Combine behavioral, textual, and emotional data sources;
- Conduct longitudinal studies on sustained engagement;
- Enhance model explainability through tailored XAI tools;
- Link predictions to adaptive instructional strategies.

In summary, progressing toward multimodal and learner-centric LA systems is essential for fully leveraging ML in higher education contexts.

3.2.2. Dropout Prediction in Digital Education with Traditional ML Models

Student attrition remains a persistent concern in digital higher education, often preceded by disengagement—behavioral, cognitive, or emotional. Predicting these early indicators is crucial for intervention. Predictive learning analytics (PLA) and machine learning (ML) are widely used to identify at-risk students before dropout. This section synthesizes findings from 23 peer-reviewed studies applying AI models across online, blended, and MOOC environments.

Random forest and SVM are the most used models, with each in nine online learning studies. Decision tree, naive Bayes, and logistic regression remain favored for their simplicity and interpretability. Ensemble methods such as random forest and boosting reduce overfitting and handle feature interactions effectively [33–35]. In MOOCs, random forest reached F1-scores above 0.88 based on daily progress [34]. Hybrid ensembles combining classification and regression enabled early risk detection before midterms [36]. During the pandemic, extra trees and logistic regression surpassed 90% specificity in asynchronous learning [37].

Neural models like LSTM and GRU outperform traditional classifiers. LSTM exceeded CNNs and MLPs from week six in STEM courses [38]; GRU achieved 98% accuracy in remedial English [39]. IOHMM and sequential logistic regression predicted dropout by week eight, outperforming SVM and logistic models [40].

Interpretable models like decision trees and logistic regression remain valuable for deployment [41,42], often as base learners in ensembles [43]. Semi-supervised models with SHAP explanations combine diverse data types [44]. Dashboards like OU analyse supported instructor monitoring and improved student outcomes [45]. Behavioral profiling through LMS logs and motivational indicators guided personalization [46,47]. Ensemble models also supported post-pandemic readiness [48], and SHAP facilitated fairness-aware predictions [49]. Self-reports on motivation were included in some models [50]. Sentiment analysis with SVM and decision trees reached over 90% accuracy [51]. Random matrix theory and community detection uncovered engagement-success links in programming education [52]. SMOTE addressed class imbalance [39]. Hybrid ensembles also explored fuzzy logic [53]. Generative models remain rare [54], and model generalizability is still limited [55].

Key applications include:

- Early warning systems in initial weeks [35,38];
- Instructor dashboards for real-time monitoring [45];
- Adaptive feedback based on predictions [35];
- Behavioral profiling using LMS and motivation data [46,47];
- Post-pandemic readiness via ensembles [48];
- Fairness-aware interventions using SHAP [49].

Challenges and opportunities: Generative AI is underused [38,54]. LMS logs dominate, while multimodal inputs are scarce [47]. Few systems operate in real time or are deployed in institutions [35,45]. Explainability tools lack integration into actionable dashboards [49]. Generalizability is limited due to narrow datasets [55].

Recommendations for future research:

- Apply generative models for feedback and simulation;
- Incorporate multimodal sources (e.g., forums, self-reports);
- Deploy real-time tools in institutional settings;
- Embed explainability in instructor-facing systems;
- Expand validation across contexts and institutions.

3.2.3. Academic Performance Prediction in Face-to-Face Classrooms

Predicting academic performance in face-to-face higher education settings remains a core objective within learning analytics (LA), primarily aiming at the early identification of students at risk. This chapter synthesizes findings from 26 studies utilizing traditional machine learning (ML) models in classroom contexts.

The most frequently employed ML algorithms include random forest (RF), support vector machines (SVM), decision trees (DT), and logistic regression, each favored due to their effectiveness with structured academic data. Specifically, RF has reliably detected underperformance through historical grades [56]. However, its predictive accuracy is notably reduced in small or homogenous student groups [57]. Conversely, SVM demonstrated strong performance with smaller datasets [58,59], though its applicability to larger classrooms remains inadequately explored. Decision trees are valued for their interpretability in guiding interventions [60] but often face challenges with imbalanced datasets [61]. Meanwhile, artificial neural networks (ANNs) effectively model complex multimodal datasets [62,63], though their inherent opacity complicates practical deployment.

Supervised learning methodologies—primarily classification and regression—dominated the reviewed studies, largely based on academic records and limited learning management system (LMS) logs. Despite the inclusion of innovative approaches such as natural language processing (NLP) for feedback analysis [62,64] and sensor-based physical interaction data [65], face-to-face contexts continue to present significant barriers due to insufficient digital interaction data for adaptive and longitudinal modeling.

Advanced analytical techniques, including concept inventories [66], gesture analytics [67], and LMS-integrated predictive models [68], contributed positively to predictive accuracy but were often limited by small sample sizes, affecting generalizability. Additionally, early assessments [69,70], demographic indicators (e.g., socioeconomic status, language proficiency, background) [71,72], and institutional data emerged as reliable predictors. Curriculum personalization emerged as a notable application. Recommendation systems were proposed based on academic pathways [73,74], aiding program design.

Key applications include:

- Performance forecasting enabling timely academic interventions [57,62,63,65,75];
- Dropout prediction identifying risk patterns for preventive actions [56,60,61,76–79];
- Personalized feedback via NLP [64];
- Curriculum personalization through predictive analytics informing tailored educational pathways [73,74,80,81].

Challenges and opportunities: Despite significant progress, key limitations persist. Generalizability remains constrained by small or imbalanced datasets and institutional variability [57,58,67]. Face-to-face contexts lack continuous digital traces, limiting adaptive and multimodal analytics. Moreover, critical variables such as motivation, emotion, and socio-economic status are often excluded despite their predictive value [60,66,71,75,76].

Addressing these issues demands not only methodological refinement but also ethical vigilance. Incorporating sensitive data raises concerns about privacy, consent, and bias. Future research must adopt privacy-preserving practices, transparent data governance, and inclusive frameworks to ensure that predictive systems are equitable, respectful of student rights, and contextually appropriate across diverse educational environments.

Recommendations for future research:

- Enhance multimodal analytics by incorporating sensor-based data (e.g., physical interactions) alongside emotional, motivational, and socio-economic variables;
- Explore advanced methodologies such as reinforcement learning and generative AI to develop dynamic, adaptive curricular recommendations;
- Prioritize explainable AI (XAI) development to enhance interpretability and facilitate educator acceptance, rigorously validating effectiveness in authentic educational environments.

3.2.4. Feedback and Performance Modeling in Hybrid Learning with ML

Machine learning is essential in hybrid learning contexts, especially for modeling academic performance and automating feedback. Artificial neural networks (ANN), random forest (RF), and support vector machines (SVM) dominate due to their effectiveness with varied educational data. ANN frequently models complex behaviors like cognitive engagement, though it faces interpretability challenges [82–84]. SVM excels at classifying high-dimensional textual data, identifying cognitively relevant content [84,85]. RF balances predictive accuracy and interpretability, analyzing variables such as peer interactions and digital distractions [83,86].

Recent use of transformer-based models like BERT has advanced analysis of emotional and behavioral signals in collaborative settings, enabling real-time personalized feedback and co-regulated learning [87].

Multimodal datasets typically combine structured data (grades, surveys) with unstructured inputs (forum posts, interaction logs, keystroke dynamics). Natural language processing (NLP) is crucial for preprocessing text, classifying engagement, and assessing discourse relevance [83,84]. Keystroke analytics reveal writing fluency and cognitive processes, while LMS logs highlight self-regulated learning patterns [88,89]. Hierarchical regression isolates cognitive, affective, and behavioral influences on student performance [90].

Explainable AI (XAI) techniques like LIME and SHAP have begun enhancing transparency and informing pedagogical decisions, yet broader integration is underdeveloped [91,92].

Key applications include:

- Cognitive engagement via discourse analysis [84];
- Analyzing distraction and peer interactions using NLP-classified discussions [83,89];
- Real-time, emotion-sensitive feedback leveraging transformer-based models [87];
- Early identification of academic risks using RF with academic and behavioral data [93–95];
- Instructional design optimization through behavioral predictors of satisfaction and performance [96].

Challenges and opportunities: Despite progress, interpretability and adaptive responsiveness remain challenges. Consistent use and pedagogical value of XAI techniques require further integration and validation [85,89,90]. Current multimodal integration mostly focuses on textual or structured inputs, limiting personalization and scalability. Incorporating audio, video, and physiological signals within diverse, multi-institutional datasets could significantly enhance analytical robustness [81,92,93]. Moreover, personalized interventions based on predictive analytics need rigorous longitudinal evaluation to confirm their educational impact [94].

Recommendations for future research:

- Integrate richer multimodal data (audio, video, physiological signals) to enhance analytical capabilities;
- Systematically apply advanced XAI techniques for increased model interpretability;
- Develop and validate adaptive, real-time feedback systems through robust longitudinal and experimental research.

Emerging technologies such as generative AI, multimodal analytics, and advanced XAI offer new opportunities to overcome current limitations in feedback and performance modeling. Their potential lies in enhancing accuracy, personalization, and pedagogical value, but this requires rigorous validation, integration into real hybrid learning contexts, and alignment with instructional goals. Future systems must evolve toward real-time, interpretable, and adaptive support that scales to diverse learner needs.

3.3. Expanding Learning Analytics Through Generative AI

The integration of generative artificial intelligence into learning analytics represents a major methodological shift in higher education. To explore this landscape, we clustered 26 GenAI-focused studies into two groups. The first cluster ($n = 14$) centers on large language models (LLMs) like GPT-4, GPT-3.5, BERT, and FLAVA, applied primarily in higher education. These studies focus on engagement analytics, adaptive learning, and ethical bias detection, often using multimodal inputs and transformer-based architectures. The second cluster ($n = 12$) features diverse approaches combining GenAI with emotion classifiers, GANs, or custom copilots, deployed across online, hybrid, and in-person contexts.

Figure 5 presents the most used GenAI models and their distribution across learning environments. GPT-4 dominates in both frequency and application scope, followed by BERT, GPT-3.5, and hybrid models (e.g., ChatGPT with clustering). Despite advances in multimodal GenAI, implementations remain largely text-based, with emphasis on automated feedback and engagement modeling.

Top 10 AI models for learning analytics with GenAI

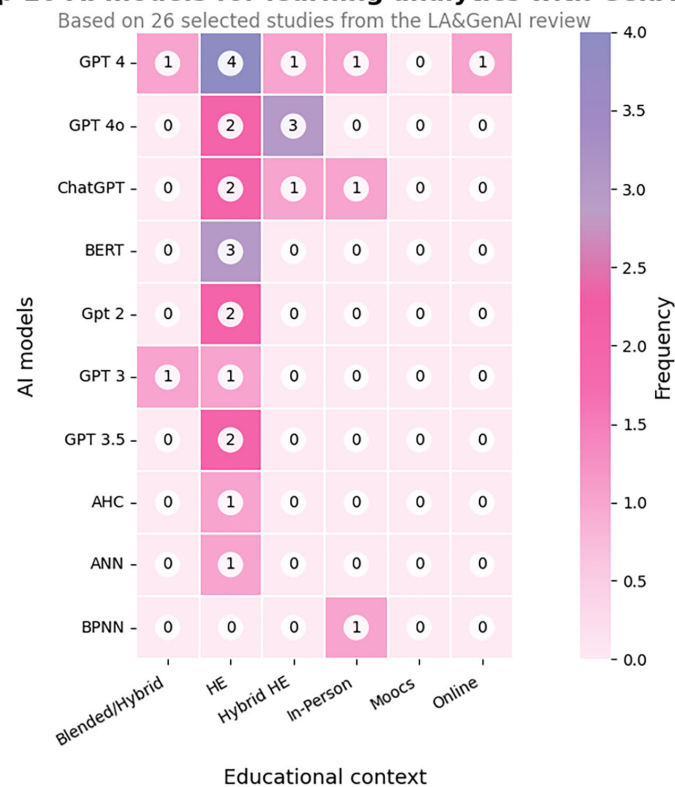


Figure 5. Top generative AI models used in learning analytics research.

These trends reflect innovation but also surface critical concerns around equity, transparency, and practical feasibility in varied educational settings.

3.3.1. Generative AI Applications for Engagement, Feedback, and Adaptation

The growing integration of generative AI (GenAI) into learning analytics (LA) underscores a methodological shift toward transformer-based models, primarily GPT-3.5, GPT-4, BERT, FLAVA, and Whisper. Frequently, these models operate within hybrid frameworks combining supervised classifiers (e.g., SVM, random forest, logistic regression) and unsupervised techniques, such as clustering and topic modeling. Among the 14 reviewed studies, GPT and BERT variants dominate, occasionally supported by recurrent neural networks (RNNs) or long short-term memory (LSTM) networks. Classical methods like decision trees and naive Bayes are occasionally used for simpler classification tasks [97]. However, the general absence of lightweight models raises scalability and equity concerns, particularly for low-resource educational contexts.

Quantitative analysis reveals GPT-4 as the most widely used model, followed by BERT and other GPT variants. Diverse methodologies, including neural networks and clustering algorithms, highlight varied applications, ranging from engagement analytics and adaptive feedback to personalized learning experiences. Large language models (LLMs) facilitate real-time reflective scaffolding [98], the detection of inclusive interactions [99], personalized feedback [100], and improved learner engagement in classroom integration contexts [101]. Some longitudinal research indicates that interaction frequency with GenAI tools enhances learner autonomy via increased social presence [102], yet rigorous empirical evidence on instructional effectiveness or scalability remains sparse.

Mixed-method approaches, integrating digital learning traces, student-generated texts, GenAI interactions, and multimodal data (audio, video, gesture), are prevalent. NLP techniques, especially sentiment analysis and topic modeling using transformers, effectively analyze textual data from learners and AI-generated content [103,104]. Clustering, combined with GPT/BERT embeddings, supports learner profiling and targeted feedback strategies [105]. Sequential models, including RNNs and LSTMs, aid the analysis of trajectories related to self-regulated learning and help-seeking behaviors [101,106].

While explainable AI (XAI) is gaining attention, its application remains limited. Fahl [107] uniquely integrates GPT-4 with semantic knowledge graphs to enhance explainability. However, broader transparency and accountability issues persist, amplifying ethical concerns such as hallucination and trustworthiness.

Key applications include:

- Formative feedback and automated assessment [100,108];
- Adaptive prompts supporting self-regulated learning [98];
- Visual analytics via GenAI-enhanced dashboards [107];
- Equity-aware collaborative learning tools [99];
- Detection of help-seeking behaviors [109];
- Adaptive content delivery in gamified or flipped learning contexts [101].

Challenges and opportunities: Although GenAI offers considerable promise, significant limitations remain. The predominant opacity of GenAI systems hinders interpretability, with rare exceptions employing explainability frameworks [107]. Additionally, the frequent misalignment with self-regulated learning (SRL) principles risks fostering metacognitive passivity, as discussed in existing critiques [110]. Real-time adaptability is another concern, as many GenAI tools provide delayed or static feedback, inadequately addressing immediate learner disengagement—though real-time adaptive scaffolding offers potential solutions [98]. The underutilization of multimodal inputs, such as audio, video, or physiological data, limits the comprehensive understanding of learner behavior and con-

text [99,103]. Lastly, weak theoretical grounding and limited alignment with established instructional frameworks diminish practical pedagogical effectiveness, highlighting the need for evidence-centered design approaches [106].

Recommendations for future research:

- Incorporate explainable AI techniques extensively to enhance transparency and trustworthiness;
- Expand multimodal GenAI applications to capture affective, cognitive, and embodied learning experiences;
- Employ ontological models to systematically structure learning progression and knowledge monitoring [107];
- Develop reflective GenAI agents supporting deep engagement and co-regulated learning dynamics.

Addressing these recommendations, along with methodological, ethical, and institutional considerations, will significantly advance the sustainable and effective integration of GenAI within higher education.

3.3.2. Technical Approaches and Emerging Trends in GenAI for Learning Analytics

The integration of generative AI (GenAI) into learning analytics (LA) is significantly reshaping methodological frameworks in higher education. Across twelve studies [111–113], large language models (LLMs), including GPT-4, GPT-4o, DistilBERT, and Gemini 1.5 Pro, have emerged as central tools for tasks such as feedback generation, affective state detection, and instructional scaffolding. Despite broad usage, most implementations remain preliminary, lacking comprehensive validation in real-world educational environments. Furthermore, the influence of instructional context on model effectiveness and learning outcomes remains underexamined.

GPT-4 variants dominate, appearing in at least eight studies, often integrated with emotion detection tools (HSEmotion), facial recognition (MTCNN), or generative adversarial networks (GANs) for enhanced multimodal analysis [114]. DistilBERT specifically excels in detecting confusion within discourse data [112]. Applications leveraging ChatGPT and GitHub Copilot have effectively captured learner strategies in coding activities [113]. Additionally, collaborative interactions supported by GPT-4 have been linked to improved hint quality and heightened critical thinking [115], while CustomGPT offers tailored insights by analyzing interactions with digital educational resources [116].

Application domains are concentrated mainly around engagement analytics, automated feedback, and risk detection. GPT-driven dashboards and conversational agents provide adaptive support and multimodal instructional scaffolding in real-time interactions [117,118]. Nevertheless, the recurrent usage of basic LLM pipelines—primarily prompt engineering combined with simple classifiers—highlights limitations in methodological innovation and adaptation to diverse instructional contexts.

The optimization of LLM outputs typically involves fine-tuning and strategic prompt design. For instance, GPT-4o has been customized to interpret natural language queries and generate educationally relevant SQL queries and visualizations [118]. Similarly, DistilBERT has successfully facilitated efficient confusion classification in MOOCs [112], while GAN-based techniques have enriched otherwise sparse datasets [114].

Some studies prioritize explainability. For instance, NLP and ChatGPT have been utilized to uncover patterns in collaboration and bias in peer-generated feedback [119]. Furthermore, GitHub Copilot interaction logs have been employed to track cognitive transitions during learning activities [113]. However, instances of feedback inaccuracies or reinforcement of misconceptions indicate the crucial role of human oversight [120].

Hybrid methodologies that combine clustering, regression, and sequential analysis with NLP techniques frequently analyze multimodal data, including LMS logs, reading behaviors, keystrokes, and emotional inputs [111,121]. Although explainable AI methods such as integrated gradients and anchors have been incorporated, their practical impact on instructional decision-making remains inadequately assessed [92,112].

Key applications include:

- Automated formative feedback in STEM disciplines through GPT4 and ChatGPT [113,115,122];
- Real-time confusion detection in online learning environments using DistilBERT [112];
- Multilingual engagement analytics leveraging bilingual prompts and chat logs [120,121];
- Bias detection and fairness evaluation in peer feedback through explainable GenAI models [119].

Challenges and opportunities: Despite promising advances, significant challenges persist. Many GenAI implementations focus predominantly on retrospective evaluation rather than proactive, real-time instructional utility. Although explainability methods are increasingly employed, their pedagogical grounding and effectiveness in supporting actual learning processes remain unclear. Ethical concerns related to inaccurate or hallucinated LLM outputs are acknowledged but insufficiently addressed through rigorous validation or expert verification [120]. Additionally, evidence supporting sustained cognitive or motivational impacts from GenAI-facilitated interactions is limited, emphasizing a critical need for longitudinal studies. Furthermore, engagement analytics tools frequently demonstrate limited transferability across educational contexts and disciplines, indicating a pressing need for scalable and adaptable solutions.

Recommendations for Future Research:

- Develop multimodal, theory-driven GenAI models integrating gaze, speech, emotions, and learner behaviors aligned explicitly with frameworks like self-regulated learning (SRL) and feedback literacy;
- Promote participatory design processes involving educators and students to define meaningful explanations, safety measures, and effective revision practices;
- Expand the evaluation criteria beyond accuracy to emphasize educational utility, fairness, epistemic validity, and ethical rigor;
- Foster scalability and transferability through adaptive models capable of continuous learning and refinement across diverse instructional contexts.

In summary, addressing foundational ethical and methodological barriers is essential to unlock the full pedagogical potential of GenAI in learning analytics.

4. Contextual Analysis of ML and GenAI in LA

The integration of artificial intelligence into learning analytics has advanced rapidly, driven by growing interest in traditional machine learning (ML) and the emerging potential of generative AI (GenAI). While ML has supported prediction and classification in education, GenAI enables real-time interaction, feedback, and adaptive content. Yet, the literature highlights persistent gaps in theoretical grounding, teacher involvement, and empirical validation. This section contrasts contributions from both domains to uncover the key trends, challenges, and future directions.

4.1. Generative AI in Learning Analytics

Recent studies emphasize GenAI's potential to transform LA from passive prediction to real-time personalization. Borah et al. [13] proposed a framework that adapts learning paths and feedback based on cognitive and emotional profiles, integrating multimodal data, contextual rules, and conversational interfaces for co-evolving learner–instructor interaction.

Complementary analyses from the LAK and EC-TEL proceedings reinforce this trend. Huang et al. [123] found RoBERTa the most accurate and explainable model for peer feedback classification, while Pishtari et al. [124] reported significant gains in instructional design quality through GenAI-generated feedback.

Yan et al. [15] mapped GenAI's integration across the LA cycle—from data augmentation to intervention—highlighting synthetic data and agent-based tools as enablers of engagement. Khosravi et al. [14] addressed ethical and pedagogical considerations, advocating codesign with educators, transparency, and models that safeguard student agency.

Qu and Yang [125] surveyed LLM applications such as ChatGPT in language and medical education, noting limited empirical validation and weak integration into instructional practice.

Despite these developments, most GenAI applications prioritize technical innovation over pedagogical grounding. Few incorporate learning theories or evaluate long-term outcomes, underscoring the need for more critically aligned, education-centered approaches.

4.2. Traditional Machine Learning in Learning Analytics

Traditional machine learning techniques have played a pivotal role in the development of learning analytics, particularly in areas such as prediction, classification, clustering, and recommendation. PeñaAyala's taxonomy [126] is frequently cited for its comprehensive organization of these functions, focusing on learner performance, engagement, and dropout analysis.

Zawacki-Richter et al. [127] found that most ML-based LA systems focus on performance prediction and early warning but often lack pedagogical alignment, limiting their educational relevance. Similarly, Renz and Hilbig [128] showed that commercial EdTech tools prioritize algorithmic efficiency over interpretability, reducing their utility for instructors.

Regionally, Salas-Pilco and Yang [129] documented promising ML applications in Latin America, such as identifying at-risk students, but the highlighted challenges included limited infrastructure, institutional support, and the need for culturally responsive models. Glandorf et al. [130] revealed that dropout prediction varies by demographic group, while Poellhuber et al. [131] improved predictive efficiency by clustering course structures in Moodle before modeling.

Buitrago-Ropero et al. [132] reframed learner data as sociopedagogical artifacts—"data, action, and service"—arguing for a shift from purely computational to sociotechnical perspectives. Baek and Doleck [133] compared educational data mining and LA, noting shared methods but differing goals: EDM focuses on algorithmic development, while LA emphasizes educational impact—though both often lack theoretical and teacher integration.

Lastly, Ley et al. [134] and Aguilar-Esteva et al. [135] called for human-centered and equity-oriented LA, promoting models that are transparent to educators and responsive to sociocultural and sustainability concerns.

4.3. Critical Synthesis and Research Gaps

Despite technological advances in both traditional ML- and GenAI-based learning analytics (LA), several persistent challenges limit their educational impact. A major concern is the lack of pedagogical integration: many systems operate independently of learning theories or instructional strategies, reducing their capacity to support meaningful learning [131]. Additionally, both ML and GenAI tools often function as opaque black boxes, hindering interpretability and diminishing educator trust in system outputs [126].

Educator involvement also remains limited. Teachers are frequently positioned as passive end users rather than active codesigners, which compromises contextual relevance and adoption [132]. Moreover, LA systems show insufficient adaptation to local constraints, including infrastructure, culture, and language, particularly in underrepresented

or resource-constrained educational settings [127]. These systemic limitations underscore the need for more inclusive and transparent LA design approaches.

To address these gaps, future research should prioritize the development of hybrid AI systems that combine the predictive strengths of ML with the interactive potential of GenAI. Such systems could better balance accuracy with personalization. Additionally, codesign practices involving both educators and learners are critical to ensure that LA tools align with real-world instructional needs. Enhancing model explainability is also essential, particularly for translating algorithmic complexity into actionable insights for educators.

Theory-driven design remains an underdeveloped area. Embedding constructs such as self-regulated learning, motivation, and feedback theory into system logic could enhance pedagogical alignment. Finally, advancing culturally responsive LA is imperative, particularly in settings with limited resources and diverse learner populations. Integrating these considerations will be key to developing LA systems that are not only innovative but also equitable, interpretable, and educationally meaningful.

These insights inform the discussion in Section 5, where we outline the broader implications for research and practice in learning analytics.

5. Discussion and Implications

This section critically synthesizes the findings of the review, addressing the two research questions and outlining their implications for educational practice, research, and institutional policy. The discussion is organized into four parts: current implementation (RQ1), potential benefits (RQ2), cross-cutting challenges, and future recommendations.

5.1. Current Implementation of ML and GenAI in Higher Education (RQ1)

The integration of machine learning and generative AI into learning analytics follows two methodological paths: the consolidation of traditional ML models and the exploratory adoption of GenAI systems.

ML-based LA systems are well-established for predicting academic performance (e.g., random forest, SVM, logistic regression), detecting dropout (e.g., LSTM, GRU, decision trees), modeling engagement (clustering, sentiment analysis), and optimizing feedback using k-means or regression. These tools operate mainly on structured LMS data—clickstreams, assessments, participation logs—and are valued for accuracy and simplicity, though they often lack alignment with classroom practice.

GenAI-based LA systems are emerging in personalized feedback (LLMs like GPT-4, BERT), emotional and cognitive engagement modeling (sentiment/discourse analysis), SRL support (tutoring agents), and real-time content scaffolding. These implementations are typically experimental, text-focused, and integrated through prompt engineering or fine-tuning. While studies—such as RoBERTa for peer feedback [123] and GenAI-enhanced instructional design [124]—show promise, most lack empirical validation, scalability, or connection to learning theory. Additionally, multimodal inputs (e.g., gaze, audio, gesture) remain underutilized.

ML-based LA systems are mature, data-driven, and prediction-focused; GenAI systems are flexible, interaction-oriented, and still experimental. Despite their complementarity, both paradigms face common limitations: fragmented implementation, weak pedagogical integration, and minimal educator involvement.

5.2. Potential Benefits of Integrating ML and GenAI into Learning Analytics (RQ2)

The integration of machine learning and generative AI into learning analytics in higher education presents benefits across pedagogical, institutional, and research domains.

Pedagogically, ML enables early risk detection, allowing timely interventions, while adaptive systems personalize feedback, pacing, and resources. GenAI tools enhance formative feedback by generating scalable, context-aware responses to student submissions. Additionally, conversational agents support metacognition and self-regulated learning (SRL) through the scaffolding of reflection, planning, and help-seeking behaviors.

Institutionally, predictive dashboards aid retention monitoring and academic advising, while LA-informed instructional design improves curriculum responsiveness. Scalable intervention systems, such as real-time alerts and adaptive prompts, enhance support in large or asynchronous courses. Cluster-based modeling of LMS data has also improved generalizability in early-warning systems, particularly when courses are grouped by structural similarity [131].

For research and innovation, multimodal learner modeling—integrating behavioral, affective, textual, and biometric data—offers a richer view of the learning process. Advances in explainable AI (XAI), including SHAP, LIME, and knowledge graphs, contribute to transparency and increase educator trust. Furthermore, hybrid LA systems that combine the predictive power of ML with the generative capabilities of GenAI enable more personalized and dynamic learning environments.

5.3. Cross-Cutting Challenges and Gaps

Despite advances in ML and GenAI for learning analytics (LA), several structural limitations persist that hinder their educational effectiveness.

A key cross-cutting gap lies in the limited grounding of LA systems—particularly those powered by GenAI—in established learning theories. For instance, several tools in the corpus provided personalized feedback using large language models, such as the explainable feedback dashboard by Afzaal et al. [91] and the fine-tuned GPT-4 system for hint revision explored by Singh et al. [115]. While both systems offer scalability and automation, they rarely aim to develop students' feedback literacy—a construct encompassing the capacity to interpret, act on, and benefit from feedback [136,137]. Moreover, the feedback remains largely unidirectional, with little scaffolding for student agency or dialogic interaction [138].

Similarly, only a subset of studies explicitly aligned with the dimensions of self-regulated learning (SRL), such as metacognitive monitoring, goal-setting, and reflection [139,140]. For example, while Dai et al. [100] focused on evaluating the quality of GPT-generated feedback for open-ended writing tasks, they did not examine how such feedback supports SRL processes or learner engagement. In contrast, Li et al. [98] integrated GenAI to deliver adaptive scaffolds based on real-time analytics of SRL processes, showing improved metacognitive strategies—yet still reported variability in learner compliance and limited generalizability.

These gaps suggest that, while ML and GenAI can technically model learner behavior, their educational impact remains limited without deliberate integration into pedagogical frameworks, such as SRL, feedback literacy, and formative assessment [141]. Additionally, many ML and GenAI models remain opaque, functioning as black boxes that impede interpretability and educator trust. While explainable AI (XAI) techniques, such as RoBERTa with LIME for peer feedback classification [123], have shown promise, they are still underused.

Educators are rarely involved as co-designers, resulting in systems poorly aligned with classroom realities. GenAI implementations are often tested in artificial or highly controlled environments. Even when benefits like improved instructional design are observed [124], few studies assess long-term impact in authentic educational settings.

Geographic disparities also persist. Research and deployment efforts are concentrated in high-income regions, limiting global representativeness and inclusivity. Latin America and other underrepresented regions remain largely absent from the discourse [129].

Finally, multimodal and real-time data, such as emotional, behavioral, or biometric streams, are rarely integrated into adaptive feedback loops, which constrains the responsiveness and depth of learner modeling.

Table 1 summarizes these cross-cutting challenges and outlines the recommended research directions to inform the development of pedagogically grounded, explainable, and context-sensitive LA systems.

Table 1. Challenges and future directions in ML and GenAI for learning Analytics.

Challenge	Description	Future Direction
Model opacity	Deep models lack transparency, limiting educational use.	Advance Explainable AI (XAI) in LA.
Post hoc modeling	Retrospective models restrict real-time interventions.	Develop online/incremental modeling and real-time feedback systems.
Modality-agnostic design	Methods are reused without context adaptation across modalities.	Create modality-aware models (e.g., sensor, LMS, multimodal).
Geographic bias	Research is concentrated in high-income regions.	Broaden global datasets and include non-Western case studies.
Limited institutional integration	Real-world deployment and constraints are rarely addressed.	Study adoption, co-design with educators, and report implementations.
Narrow evaluation metrics	Accuracy often outweighs pedagogical value in evaluations.	Integrate educational impact and instructional alignment.
Ethical blind spots	Fairness, consent, and bias remain underexplored.	Apply ethical audits and promote fairness-aware ML and data literacy.
Underdeveloped GenAI use	GenAI lacks pedagogical grounding and deep integration into LA.	Explore adaptive, generative, and co-creative GenAI applications.

5.4. Implications and Recommendations

5.4.1. Implications of Generative AI for Engagement, Feedback, and Adaptation

Instructors should be actively involved in the co-design of LA systems to ensure alignment with pedagogical goals. Hybrid models that combine ML's predictive strength with GenAI's adaptive feedback capabilities offer promise for more responsive learning environments. Embedding explainability into system interfaces can enhance transparency, trust, and interpretability for both educators and learners.

5.4.2. Research

Future studies should move beyond predictive accuracy and assess actual learning outcomes through longitudinal and mixed methods designs. Aligning model architectures with learning theories, such as SRL, motivation, and engagement, can improve pedagogical relevance. Additionally, the instructional impact of GenAI tools on reflection, autonomy, and knowledge construction warrants deeper exploration.

5.4.3. Institutions and Policy

Institutions must establish ethical frameworks and data governance policies to guide the responsible use of LA and GenAI. Bridging digital gaps in under-resourced contexts is essential to prevent widening educational inequities. Professional development in AI and data literacy should be prioritized for educators, advisors, and policymakers to support informed implementation.

5.5. Looking Forward: Toward Human-Centered, Hybrid Learning Analytics

This review identifies a methodological shift in learning analytics (LA), from retrospective analysis to adaptive, generative, and participatory systems. While ML contributes structure and predictive rigor, GenAI adds adaptability and interactive potential. Their convergence—when guided by educational theory and ethical principles—can foster LA tools that are transparent, equitable, and pedagogically aligned.

Achieving this vision requires moving beyond technical prototypes toward empirical validation, contextual deployment, and active stakeholder collaboration. Future systems must be co-designed, responsive to real educational settings, and focused on empowering both educators and learners.

Ultimately, the future of LA lies not only in algorithmic advancement but in designing systems that serve learning with purpose, transparency, and equity at their core.

6. Study Limitations

Achieving the transformative potential of ML and GenAI in education requires moving beyond technical prototypes toward empirical validation, contextual deployment, and active stakeholder collaboration. Future systems must be co-designed, responsive to authentic educational settings, and focused on empowering both educators and learners.

First, all screening stages (titles/abstracts and full texts) were performed by a single reviewer. While eligibility criteria were clearly defined, this introduces potential selection bias.

Second, although a structured quality appraisal was conducted using the Mixed Methods Appraisal Tool (MMAT), the high methodological heterogeneity across studies limited the possibility of assigning comparable quality scores or excluding low-quality papers. Instead, the appraisal was used descriptively to inform interpretations and increase transparency.

Third, restricting the search to English-language publications may have excluded relevant studies from non-English-speaking regions, potentially limiting the cultural and geographical representativeness of the synthesis. Expanding future reviews to include multilingual sources would enhance global inclusivity.

Finally, although the main corpus focused on peer-reviewed journal articles indexed in major databases, an exploratory scan of LAK, L@S, and EC-TEL proceedings (2018–2025) was conducted. While these records were not formally included in the PRISMA dataset, selected studies were referenced to triangulate findings and highlight emerging developments.

Acknowledging these limitations is key to contextualizing the scope of this review and guiding future research toward greater methodological rigor, inclusiveness, and practical relevance.

7. Conclusions

This review offers a critical synthesis of how machine learning (ML) and generative AI (GenAI) are shaping learning analytics (LA) in higher education. Based on 101 empirical studies (2018–2025), along with insights from recent conferences, it highlights both methodological advances and persistent challenges across academic, technological, and pedagogical dimensions.

Traditional ML models remain central to LA, particularly for performance prediction, dropout detection, and engagement analysis. While robust and interpretable, these models often rely on retrospective, mono-modal data and show limited integration with pedagogical practices. In contrast, GenAI systems—particularly those using large language models (LLMs) like GPT-4—offer promising innovations in personalized feedback and affective modeling, yet remain largely experimental, with minimal real-world validation or theoretical grounding.

Recent efforts to enhance explainability, leverage clustering, and support instructional design mark progress, but critical gaps persist. These include limited use of multimodal

and real-time data, weak ethical oversight, minimal educator involvement, and a lack of evidence for sustained impact on learning.

Rather than a paradigm shift, the current trends reveal a transitional phase in which ML and GenAI approaches coexist. This duality calls for hybrid, human-centered systems that combine predictive accuracy with adaptive feedback, grounded in learning theory and responsive to educational context.

Future progress requires a shift from technical performance to pedagogical impact—from opaque experimentation to scalable, explainable, and inclusive implementations. Advancing LA in line with Sustainable Development Goals 4.3 and 4.4 depends not solely on technological innovation, but on institutional support, educator engagement, and critical reflection.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/app15158679/s1>, <https://doi.org/10.5281/zenodo.15233231>: Table S1: Summary of Queries Used in Databases; <https://doi.org/10.5281/zenodo.16422602>: PRISMA Checklist. Available and <https://doi.org/10.5281/zenodo.16416487>: MMAT Quality Appraisal at Zenodo.

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Abbreviations

The following abbreviations are used in this manuscript:

AHC	Agglomerative Hierarchical Clustering
ANN	Artificial Neural Network
BERT	Bidirectional Encoder Representations from Transformers
DT	Decision Tree
GenAI	Generative Artificial Intelligence
GPT	Generative Pre-trained Transformer
GRU	Gated Recurrent Unit
LA	Learning Analytics
LIME	Local Interpretable Model-Agnostic Explanations
LLM	Large Language Model
LMS	Learning Management System
LSTM	Long Short-Term Memory
ML	Machine Learning
MOOC	Massive Open Online Course
NLP	Natural Language Processing

PLA	Predictive Learning Analytics
RF	Random Forest
SHAP	SHapley Additive exPlanations
SRL	Self-Regulated Learning
SVM	Support Vector Machine
XAI	Explainable Artificial Intelligence

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